

SIMON V3 Research - File Inventory for Mentor Review

Scripts, outputs, charts and proposed paper figures for each experiment

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1. Architecture Comparison: Proving Analytic Force Direction Is Necessary

Controlled comparison between SIMON (analytic direction) and a new Vector NN baseline (learned direction). Both models use identical simulation conditions, same timestep, same initial conditions, equivalent safety gating. Experiment run from: C:\Aarush\Physics\training\

Scripts / Input Files

File name	Role and description
train_vector_nn.py	Trains the Vector NN baseline model. Architecture: 7->32->32->32->3 (SiLU). Inputs: [rx/r, ry/r, rz/r, log(r_soft), log(r), log(mi), log(mj)]. Output: unit direction vector. 20,000 epochs, AdamW with cosine annealing. Run once to produce vector_nn.pt.
train_pair_correction_new.py	Original SIMON training script (unchanged from V2). Architecture: 3->32->32->32->1. Predicts scalar correction log(c). Already trained; included for reference.
compare_models.py	Runs ias15, SIMON, and Vector NN through identical 100-year leapfrog simulations at dt=0.04 yr. Computes divergence rate lambda for each model and generates all output charts. Command: python compare_models.py
pair_correction_nn.pt	SIMON model weights (seed 42, 2,273 params, 13.3 KB). Pre-existing from V2.
vector_nn.pt	Vector NN model weights (2,467 params, 13.7 KB). Generated by train_vector_nn.py.

Output Files - C:\Aarush\Physics\training\comparison_out\

File name	Description
divergence_3model.png	Log-scale RMS position error vs ias15 over 100 years for SIMON (green) and Vector NN (red). Shows lambda values and the Vector NN ejection event. PROPOSED FOR PAPER.
trajectory_3model.png	XY trajectory overlay for all 3 bodies: ias15 (blue), SIMON (green dashed), Vector NN (red dotted). Shows ejection trajectory of Vector NN bodies. PROPOSED FOR PAPER.
timeseries_3model.png	Position x(t) and y(t) for each body over 100 years, all three models. Shows Vector NN ejection diverging from bounded behaviour around year 35.
summary_comparison.txt	Quantitative results table: lambda, final RMS, speedup, NN fraction for each model. Key result: SIMON lambda=0.1655/yr RMS=1.54 AU bounded; Vector NN lambda=0.1656/yr RMS=83.0 AU ejected.

Robustness Verification Files - C:\Aarush\Physics\training\

File name	Description
comparison_out_seed456\	Full comparison re-run using worst-seed model (seed 456, highest training MSE). Confirms identical lambda=0.1655 and ejection result. Same conclusion.
comparison_out_seed1000\	Full comparison re-run using seed 1000 model. Confirms identical results.

Charts Proposed for Final Paper

Chart	File	What it shows
Divergence rate comparison	divergence_3model.png	Log-scale RMS error over 100 yr. SIMON stays bounded, Vector NN ejects at year 35. Identical lambda during bounded phase proves same chaotic physics; ejection proves direction constraint is necessary.
Trajectory overlay (3 bodies)	trajectory_3model.png	XY trajectories showing SIMON/ias15 bounded overlap vs Vector NN ejection extending to 125+ AU.

2. Multi-Initial-Condition Evaluation: Demonstrating Generalisation

SIMON evaluated on 4 qualitatively different orbital configurations to test generalisation beyond the single V2 initial condition. Includes physical instability verification for the ejecting IC using ias15 as reference.

Scripts - C:\Aarush\Physics\training\

File name	Role and description
multi_ic_eval.py	Runs SIMON and ias15 on 4 initial conditions: IC1 default (V2), IC2 near-equal mass, IC3 tight inner pair, IC4 hierarchical. Command: python multi_ic_eval.py
check_ic2.py	Standalone verification script: runs ias15 alone on IC2 (near-equal mass) to confirm physical instability. ias15 ejects a body to 58.9 AU, proving IC2 ejection is physical not numerical. Command: python check_ic2.py
pair_correction_nn.pt	SIMON model used for all 4 ICs.

Output Files - C:\Aarush\Physics\training\multi_ic_out\

File name	Description
multi_ic_divergence.png	4-panel divergence curve figure, one panel per IC. Shows lambda and bounded/ejection status. IC2 shows ejection trajectory.
multi_ic_results.png	3-panel bar chart: lambda, final RMS, and speedup for all 4 ICs. IC2 shown with hatching to indicate physical instability. PROPOSED FOR PAPER.
multi_ic_trajectories.png	Trajectory overlay (Body 1 shown for each IC): ias15 vs SIMON comparison across all 4 configurations.
multi_ic_summary.txt	Quantitative table: lambda, final RMS, speedup, NN fraction, bounded status for each IC. Mean lambda for bounded ICs = $0.128 \pm 0.028/\text{yr}$.

Charts Proposed for Final Paper

Chart	File	What it shows
Multi-IC summary bar charts	multi_ic_results.png	3 panels: lambda, final RMS, speedup across 4 ICs. Demonstrates generalisation. IC2 hatched to distinguish physical instability from model failure.

3. Energy Conservation Tracking: A Second Independent Stability Metric

Fractional energy drift dE/E_0 measured over 100 years for ias15 and SIMON across 3 stable ICs (IC1 default, IC3 tight, IC4 hierarchical). Fully self-contained script - no imports needed.

Scripts - C:\Aarush\Physics\training\

File name	Role and description
energy_drift_eval.py	Self-contained: runs ias15 and SIMON for IC1, IC3, IC4. Computes total mechanical energy $E(t) = KE + PE$ at each of 5,000 recorded timesteps. Plots fractional drift dE/E_0 over 100 years. Command: <code>python energy_drift_eval.py</code>
pair_correction_nn.pt	SIMON model used for all 3 ICs.

Output Files - C:\Aarush\Physics\training\energy_drift_out\

File name	Description
energy_drift_ic1_default.png	Detailed 2-panel plot for default IC: log-scale $ dE/E_0 $ vs time (left) and linear fractional drift % (right). Shows ias15 flat at $\sim 1e-14$ and SIMON oscillating around 1.4%.
energy_drift_all_ics.png	3-panel figure: one panel per IC showing ias15 (flat) and SIMON (drift) side by side over 100 years. PROPOSED FOR PAPER.
energy_drift_combined.png	Bar chart summary: max drift per IC. NOTE: the SIMON/ias15 ratio column (10^{12} to 10^{14}) should NOT be used in paper - report absolute percentages only (IC1=1.39%, IC3=0.67%, IC4=13.77%).
energy_drift_summary.txt	Quantitative table: IC, E_0 , ias15 max drift, SIMON max drift, ratio. Key paper numbers: IC1=1.39%, IC3=0.67%, IC4=13.77% (IC4 explained by small $ E_0 $ and leapfrog integrator, not NN failure).

Charts Proposed for Final Paper

Chart	File	What it shows
Energy drift across 3 ICs	energy_drift_all_ics.png	ias15 at float64 machine precision vs SIMON leapfrog drift. IC3 and IC4 have zero NN invocations yet still show drift - proving energy drift is integrator artefact, not NN-caused.

4. Adaptive Sub-Stepping: Quantifying the Most Impactful Engineering Decision

Before/after comparison using actual hardware runs already stored in two separate result folders. Comparison script reads both `perf_summary` files and generates consolidated figures.

Scripts - C:\Aarush\Physics\Results for paper\

File name	Role and description
adaptive_comparison.py	Reads <code>perf_summary_T100_before_adaptive.txt</code> and <code>perf_summary_T100_after_adaptive.txt</code> from the two result folders. Generates side-by-side comparison figures and copies all existing PNGs into one consolidated output folder. Command: <code>python adaptive_comparison.py</code>

Input Data Files (pre-existing hardware results, read-only)

File / folder	Description
<i>set 2_before_adaptive\hybrid_eval_out_v3\perf_summary_T100_before_adaptive.txt</i>	Performance summary from before-adaptive run. Key numbers: lambda=0.2508/yr, worst error=161.7 AU, step time=17.9 us.
<i>set 3_after_adaptive\hybrid_eval_out_v3\perf_summary_T100_after_adaptive.txt</i>	Performance summary from after-adaptive run. Key numbers: lambda=0.1562/yr, worst error=1.54 AU, step time=24.3 us.
<i>set 2_before_adaptive\hybrid_eval_out_v3*.png</i>	5 original PNGs from before-adaptive run: divergence, speed-accuracy, trajectory, timeseries, comp_cost.
<i>set 3_after_adaptive\hybrid_eval_out_v3*.png</i>	5 original PNGs from after-adaptive run.

Output Files - C:\Aarush\Physics\Results for paper\output\

File name	Description
<i>adaptive_comparison_combined.png</i>	3-panel figure: (a) error vs dt before/after with 105x annotation, (b) lambda bar chart showing 38% improvement, (c) per-step time bar showing +35% overhead. PROPOSED FOR PAPER.
<i>adaptive_error_vs_dt.png</i>	Standalone larger version of error vs dt chart with 105x improvement annotation.
<i>adaptive_divergence_slope.png</i>	Bar chart: lambda before (0.251/yr) vs after (0.156/yr), -38% reduction.
<i>adaptive_step_time.png</i>	Bar chart: per-step time 17.9 us vs 24.3 us, +35% overhead.
<i>adaptive_comparison_table.txt</i>	Before/after table with all key numbers for paper section.
<i>model_vs_rebound_divergence_T100_before/after_adaptive.png</i>	Original divergence plots copied from both result folders. Before: shows discontinuous ejection jump at year ~30. After: smooth exponential growth, no discontinuities. Striking visual contrast.
<i>speed_accuracy_frontier_T100_before/after_adaptive.png</i>	Speed-accuracy frontier plots. Before: 162 AU outlier at dt=0.01. After: clean cluster within [0.847, 1.543] AU, no outliers.

Charts Proposed for Final Paper

Chart	File	What it shows
3-panel before/after comparison	<i>adaptive_comparison_combined.png</i>	Primary figure for new Section 4.4. Shows 105x error reduction, 38% lambda improvement, 35% overhead - complete cost/benefit picture.
Before and after divergence curves	<i>divergence_T100_before/after.png</i>	Side-by-side: ejection discontinuity (before) vs smooth exponential growth (after). Visual proof of the stability transition.

5. Training Seed Robustness: Confirming Reproducibility

SIMON retrained 5 times with different random seeds (same data). Worst-seed model then run through full 100-year simulation to confirm identical downstream performance despite variation in training loss.

Scripts - C:\Aarush\Physics\training\

File name	Role and description
seed_robustness.py	Trains SIMON 5 times (seeds 42, 123, 456, 789, 1000). Same data seed (42) for all - tests model weight initialisation sensitivity only. Saves 5 .pt files plus summary. Command: python seed_robustness.py
compare_models.py	Reused with --simon_model and --out_dir flags to run worst seed (456) and second seed (1000) through full Vector NN comparison simulation.

Output Files - C:\Aarush\Physics\training\

File name	Description
pair_correction_nn_seed42.pt	SIMON model, seed 42 (best val MSE: 1.113e-05).
pair_correction_nn_seed123.pt	SIMON model, seed 123 (val MSE: 1.723e-05).
pair_correction_nn_seed456.pt	SIMON model, seed 456 (worst val MSE: 1.754e-05).
pair_correction_nn_seed789.pt	SIMON model, seed 789 (val MSE: 1.683e-05).
pair_correction_nn_seed1000.pt	SIMON model, seed 1000 (val MSE: 1.430e-05).
seed_robustness_results.txt	Summary: best val MSE per seed, mean +/- std (1.541 +/- 0.243 x 10 ⁻⁵), CV=15.8%. All seeds produce identical simulation results.
seed_robustness_results.png	2-panel figure: all 5 training curves overlaid (left) + bar chart of best MSE per seed (right). PROPOSED FOR PAPER.
comparison_out_seed456\summary_comparison.txt	Simulation results for worst seed: lambda=0.1655/yr, RMS=1.23 AU, bounded=YES. Identical to seed 42 result. Confirms robustness.
comparison_out_seed1000\summary_comparison.txt	Simulation results for seed 1000: identical to seed 42.

Charts Proposed for Final Paper

Chart	File	What it shows
Seed robustness training curves + bar chart	seed_robustness_results.png	Shows 15.8% CV in training MSE across seeds (visible spread in bar chart) while simulation outcomes are identical. Supports reproducibility claim.

6. Reviewer Feedback Response: Resolving Five Identified Issues

Analysis of 5 issues from an external reviewer against the V2 paper. A targeted NN ablation experiment (no_nn_ablation.py) was run to address Issue 1 with direct experimental evidence. Issues 2-5 require paper text corrections only.

Scripts - C:\Aarush\Physics\training\

File name	Role and description
no_nn_ablation.py	Runs SIMON vs No-NN baseline (pure softened gravity, correction factor c=1 always) through identical 100-year simulations. Tests whether the NN correction is essential at close encounters. Self-contained, simulation loop aligned with compare_models.py. Command: python no_nn_ablation.py [output -> comparison_out_no_nn\]

File name	Role and description
pair_correction_nn.pt	SIMON model used for both SIMON and No-NN modes. In No-NN mode the NN forward pass is simply skipped; softened gravity used directly.

Output Files - C:\Aarush\Physics\training\comparison_out_no_nn\

File name	Description
divergence_no_nn.png	SIMON vs No-NN divergence curves over 100 years. Both models stay bounded; curves nearly identical. Confirms NN adds accuracy, not primary stability. PROPOSED FOR PAPER.
trajectory_no_nn.png	XY trajectory overlay for all 3 bodies: ias15, SIMON (with NN), No-NN baseline.
timeseries_no_nn.png	Position x(t) and y(t) for all 3 bodies comparing SIMON and No-NN over 100 years.
summary_no_nn.txt	Quantitative results: SIMON lambda=0.1655 RMS=1.54 bounded=YES; No-NN lambda=0.1655 RMS=1.67 bounded=YES. Verdict: NN provides 8.5% accuracy improvement; adaptive sub-stepping is the primary stability mechanism (confirmed by Run A: adaptive OFF, NN ON -> ejection 162 AU).

Three-Run Ablation Framework (combining Steps 4 and 6 results)

Run	Design and result - source file
Run A: adaptive OFF, NN ON	Body ejection to 161.7 AU. Source: perf_summary_T100_before_adaptive.txt (from Experiment 4). Proves adaptive stepping is essential.
Run C: adaptive ON, NN OFF	Bounded, RMS=1.67 AU. Source: summary_no_nn.txt (this experiment). Proves adaptive stepping alone is sufficient for stability.
Run D: adaptive ON, NN ON	Bounded, RMS=1.54 AU. Source: perf_summary_T100_after_adaptive.txt (from Experiment 4). SIMON baseline - best result.

Charts Proposed for Final Paper

Chart	File	What it shows
SIMON vs No-NN divergence	divergence_no_nn.png	Both bounded; 8.5% larger RMS without NN. Combined with Run A result, establishes full 3-run ablation framework for the paper's new Section 4.6.

All scripts run on NVIDIA GeForce RTX 5050, CUDA 13.0, Python 3.10.11, PyTorch 2.9.0+cu130, REBOUND. Detailed mentor reports for each experiment are available separately.